

Lissajous Figures

When a particle is influenced simultaneously by two simple harmonic motions at right angles to each other, the resultant motion of the particle traces a curve. These curves are called a Lissajous figures. The shape of the curve depends on the time period, phase difference and the amplitude of the two constituent vibrations. Lissajous figures are helpful in determining the ratio of the time periods of two vibrations and to compare the frequencies of two tuning forks.

Composition of two simple harmonic motions in a straight line having same frequency but different phase and amplitude

Let the two simple harmonic motions acting on the particle be represented by the equations

$$y_1 = a_1 \sin(\omega t + \alpha_1) \dots\dots\dots(1)$$

$$y_2 = a_2 \sin(\omega t + \alpha_2) \dots\dots\dots(2)$$

where, y_1 and y_2 are the displacements of a particle due to the two vibrations a_1 and a_2 are the amplitudes of the two vibrations and α_1 and α_2 are the epoch angles. Here the two vibrations are assumed to be of the same frequency and hence ω is the same for both. The resultant displacement y of the particle is given by

$$\begin{aligned} y &= y_1 + y_2 \\ &= a_1 \sin(\omega t + \alpha_1) + a_2 \sin(\omega t + \alpha_2) \\ &= a_1 (\sin\omega t \cos\alpha_1 + \cos\omega t \sin\alpha_1) + a_2 (\sin\omega t \cos\alpha_2 + \cos\omega t \sin\alpha_2) \\ &= (a_1 \cos\alpha_1 + a_2 \cos\alpha_2) \sin\omega t + (a_1 \sin\alpha_1 + a_2 \sin\alpha_2) \cos\omega t \dots\dots\dots(3) \end{aligned}$$

Since the amplitudes a_1 and a_2 and the angles α_1 and α_2 are constant, the coefficients of $\sin\omega t$ and $\cos\omega t$ in equation (3) can be substituted by $A \cos \phi$ and $A \sin \phi$

$$A \cos \phi = a_1 \cos\alpha_1 + a_2 \cos\alpha_2 \dots\dots\dots(4)$$

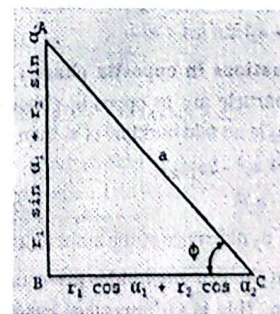
$$A \sin \phi = a_1 \sin\alpha_1 + a_2 \sin\alpha_2 \dots\dots\dots(5)$$

Then, squaring and adding equations (4) and (5)

$$\begin{aligned} A^2 (\sin^2\phi + \cos^2\phi) &= a_1^2 (\sin^2\alpha_1 + \cos^2\alpha_1) + a_2^2 (\sin^2\alpha_2 + \cos^2\alpha_2) \\ &\quad + 2a_1a_2 [\cos\alpha_1\cos\alpha_2 + \sin\alpha_1\sin\alpha_2] \end{aligned}$$

$$A^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos(\alpha_1 - \alpha_2) \dots\dots\dots(6)$$

Dividing equation (5) by (4)



$$\tan \phi = \frac{a_1 \sin \alpha_1 + a_2 \sin \alpha_2}{a_1 \cos \alpha_1 + a_2 \cos \alpha_2} \dots\dots\dots(7)$$

From equation (3), the resultant displacement can be written as

$$y = A \cos \phi \sin \omega t + A \sin \phi \cos \omega t$$

$$y = A \sin (\omega t + \phi) \dots\dots\dots(8)$$

Equation (8) is similar to the original equation (1) and (2), where the amplitude of the resultant vibration is A and epoch angle is ϕ . The time period of the resultant vibration is same as the original vibrations. Thus, the resultant vibration is also simple harmonic motion of the same period and acting in the same straight line with a resultant amplitude A and epoch angle ϕ .

Special cases:

- (a) Same phase: if the two simple harmonic vibrations acting on the particle are in same phase, then $\alpha_1 = \alpha_2 = \alpha$ i.e., if the phase difference $(\alpha_1 - \alpha_2)$ is zero or an integral multiple of 2π , then

$$A = a_1 + a_2 = \text{sum of the amplitudes and } \phi = \alpha$$

or, $y = (a_1 + a_2) \sin (\omega t + \alpha) \dots\dots\dots(9)$

- (b) Opposite phase: if the two simple harmonic vibrations on the particle are in opposite phase, i.e., if the phase difference $(\alpha_1 - \alpha_2)$ is an odd of π , then

$$A = a_1 - a_2 = \text{difference of the amplitudes}$$

If, in addition, the two amplitudes of the individual vibrations are equal, i.e., $a_1 = a_2 = a$ (say), then for the same phase condition $A = 2a$ and $A^2 = 4a^2$, while in the case of opposite phase $A = 0$ (i.e., the resultant vibration is zero).

Composition of two simple harmonic motions at right angles to each other having equal frequency but differing in phase and amplitude

Let us consider two simple harmonic motions of the same frequency (i.e., same time period) but of amplitude a and b and having their vibrations mutually perpendicular to one another. If α is the phase difference between two vibrations, then their equation can be written as

$$x = a \sin (\omega t + \alpha) \dots\dots\dots(10)$$

and $y = b \sin \omega t \dots\dots\dots(11)$

represent the displacements of a particle along the X- and Y-axes.

From equation (11), $\sin \omega t = \frac{y}{b}$

$$\cos \omega t = \sqrt{1 - \frac{y^2}{b^2}}$$

From equation (10), $\frac{x}{a} = [\sin \omega t \cos \alpha + \cos \omega t \sin \alpha]$ (12)

Substituting the values of $\sin \omega t$ and $\cos \omega t$ in equation (12)

$$\frac{x}{a} = \left[\frac{y}{b} \cos \alpha + \sqrt{1 - \frac{y^2}{b^2}} \sin \alpha \right]$$

or, $\frac{x}{a} - \frac{y}{b} \cos \alpha = \sqrt{1 - \frac{y^2}{b^2}} \sin \alpha$

Squaring both sides,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \cos^2 \alpha - \frac{2xy}{ab} \cos \alpha = \left(1 - \frac{y^2}{b^2} \right) \sin^2 \alpha$$

or, $\frac{x^2}{a^2} + \frac{y^2}{b^2} [\sin^2 \alpha + \cos^2 \alpha] - \frac{2xy}{ab} \cos \alpha = \sin^2 \alpha$

or, $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} \cos \alpha = \sin^2 \alpha$ (13)

This represents the general equation of an ellipse. Thus, due to the superimposition of two simple harmonic vibrations, the displacement of the particle will be along a curve (Fig. 1) given by equation (13). The resultant vibration of the particle will depend upon the value of α . Fig. 1 represents the resultant vibration for values of α changing from 0 to 2π .

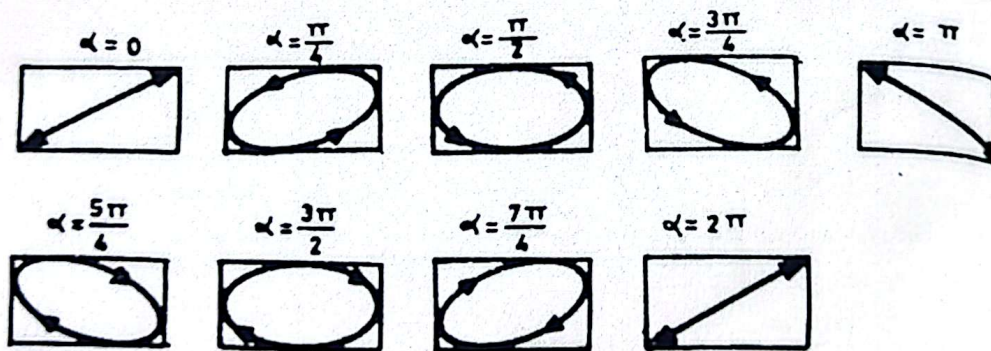


Fig. 1

Special cases:

(i) If $\alpha = 0$ or 2π

$$\cos \alpha = 1, \sin \alpha = 0$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} = 0$$

or, $\frac{x}{a} - \frac{y}{b} = 0$

$$\text{or, } y = \frac{b}{a} x$$

This represents the equation of the straight line BD (Fig. 2) i.e., the particle vibrates simple harmonically along the line DB.

$$(ii) \quad \text{If } \alpha = \pi, \sin \alpha = 0, \cos \alpha = -1$$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{2xy}{ab} = 0$$

$$\text{or, } \frac{x}{a} + \frac{y}{b} = 0$$

$$\text{or, } y = -\frac{b}{a} x$$

This represents the equation of the straight line AC (Fig. 2).

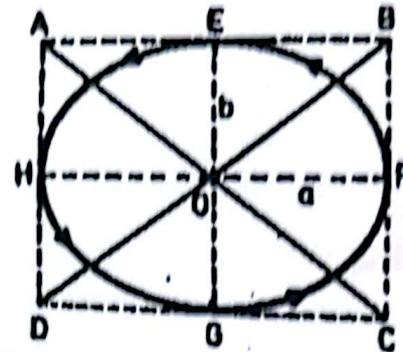


Fig. 2

$$(iii) \quad \text{If } \alpha = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}, \sin \alpha = 1, \cos \alpha = 0$$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

This represents the equation of the ellipse EHGF (Fig. 2) with a and b as the semi-major and semi-minor axes.

$$(iv) \quad \text{If } \alpha = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \text{ and } a = b$$

$$\text{then, } \frac{x^2}{a^2} + \frac{y^2}{a^2} = 1$$

$$\text{or, } x^2 + y^2 = a^2$$

This represents the equation of a circle (Fig. 3).

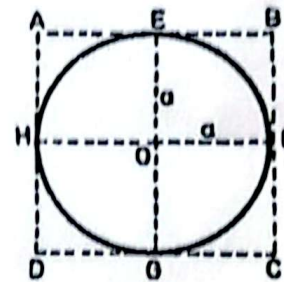
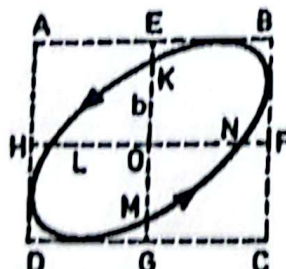


Fig. 3

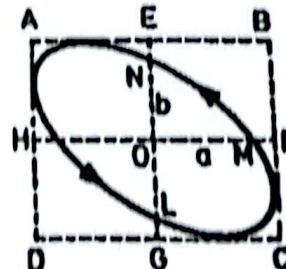
$$(v) \quad \text{If } \alpha = \frac{\pi}{4} \text{ or } \frac{7\pi}{4}$$

The resultant vibration is an oblique ellipse KLMN as shown in Fig. 4(i), the equation of the oblique ellipse being

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{\sqrt{2}xy}{ab} = \frac{1}{2} \left[\text{for } \alpha = \frac{\pi}{4} \right]$$



(i)



(ii)

Fig. 4

On the other hand, if $\alpha = \frac{3\pi}{4}$ or $\frac{5\pi}{4}$

The resultant vibration is again an oblique ellipse KLMN as shown in Fig. 4 (ii). The cycle of changes is repeated after every time period. The equation of the oblique ellipse becomes in this case,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} \cos 135^\circ = \sin^2 135^\circ \left[\text{for } \alpha = \frac{3\pi}{4} = 135^\circ \right]$$

$$\text{or, } \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} \left(-\frac{1}{\sqrt{2}} \right) = \left(-\frac{1}{\sqrt{2}} \right)^2$$

$$\text{or, } \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{\sqrt{2}xy}{ab} = \frac{1}{2}$$

Composition of two simple harmonic motions at right angles to each other and having time periods in the ratio 1:2

Let the equations of the two simple harmonic motions be

$$x = a \sin (2 \omega t + \alpha) \dots\dots\dots(14)$$

$$\text{and } y = b \sin \omega t \dots\dots\dots(15)$$

Here, a is the amplitude for the motion along the X-axis and b is the amplitude for the motion along the Y-axis. The phase difference between the two vibrations is α .

From equation (15)

$$\frac{y}{b} = \sin \omega t \dots\dots\dots(16)$$

$$\text{and } \cos \omega t = \sqrt{1 - \sin^2 \omega t}$$

$$\text{or, } \cos \omega t = \sqrt{1 - \frac{y^2}{b^2}}$$

From equation (14)

$$\begin{aligned} \frac{x}{a} &= \sin (2 \omega t + \alpha) \\ &= \sin 2\omega t \cos \alpha + \cos 2\omega t \sin \alpha \\ &= 2 \sin \omega t \cos \omega t \cos \alpha + (1 - 2 \sin^2 \omega t) \sin \alpha \end{aligned}$$

Substituting the values of $\sin \omega t$ and $\cos \omega t$.

$$\frac{x}{a} = 2 \cdot \frac{y}{b} \cdot \sqrt{1 - \frac{y^2}{b^2}} \cos \alpha + \left(1 - 2 \frac{y^2}{b^2} \right) \sin \alpha$$

$$\text{or, } \left[\frac{x}{a} - \left(1 - \frac{2y^2}{b^2} \right) \sin \alpha \right] = \frac{2y}{b} \cos \alpha \sqrt{1 - \frac{y^2}{b^2}}$$

$$\text{or, } \left[\left(\frac{x}{a} - \sin \alpha \right) + \frac{2y^2}{b^2} \sin \alpha \right] = \frac{2y \cos \alpha}{b} \sqrt{1 - \frac{y^2}{b^2}}$$

Squaring both sides

$$\begin{aligned} & \left(\frac{x}{a} - \sin \alpha\right)^2 + \frac{4y^4}{b^4} \sin^2 \alpha + 2 \left(\frac{x}{a} - \sin \alpha\right) \frac{2y^2}{b^2} \sin \alpha = \frac{4y^2 \cos^2 \alpha}{b^2} \left(1 - \frac{y^2}{b^2}\right) \\ \text{or, } & \left(\frac{x}{a} - \sin \alpha\right)^2 + \frac{4y^4}{b^4} (\sin^2 \alpha + \cos^2 \alpha) - \frac{4y^2}{b^2} (\sin^2 \alpha + \cos^2 \alpha) + \frac{4y^2}{b^2} \frac{x}{a} \sin \alpha = 0 \\ \text{or, } & \left(\frac{x}{a} - \sin \alpha\right)^2 + \frac{4y^4}{b^4} - \frac{4y^2}{b^2} + \frac{4y^2}{b^2} \frac{x}{a} \sin \alpha = 0 \\ \text{or, } & \left(\frac{x}{a} - \sin \alpha\right)^2 + \frac{4y^2}{b^2} \left[\frac{y^2}{b^2} + \frac{x}{a} \sin \alpha - 1\right] = 0 \quad \dots\dots\dots(17) \end{aligned}$$

Equation (17) represents the general equation of a curve having two loops. The resultant motion of the particle for different values of α is given in Fig. 5. For a phase difference of 0, π and 2π , the resultant motion gives the figure of eight.

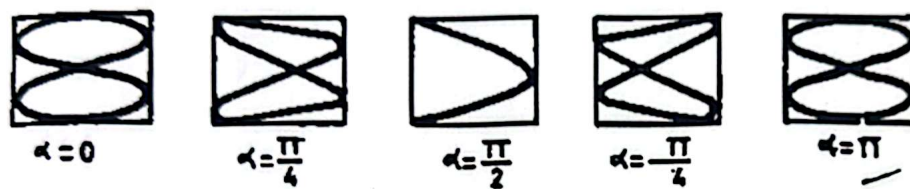


Fig. 5

Special cases:

(i) When $\alpha = 0, \pi, 2\pi$, etc., $\sin \alpha = 0$

From equation (17)

$$\frac{x^2}{a^2} + \frac{4y^2}{b^2} \left(\frac{y^2}{b^2} - 1\right) = 0$$

This equation represents the figure of eight and has two loops.

(ii) When $\alpha = \frac{\pi}{2}$, $\sin \alpha = +1$

From equation (17)

$$\begin{aligned} & \left(\frac{x}{a} - 1\right)^2 + \frac{4y^2}{b^2} \left(\frac{y^2}{b^2} + \frac{x}{a} - 1\right) = 0 \\ \text{or, } & \left(\frac{x}{a} - 1\right)^2 + \frac{4y^2}{b^2} \left(\frac{x}{a} - 1\right) + \frac{4y^4}{b^4} = 0 \\ \text{or, } & \left[\left(\frac{x}{a} - 1\right) + \frac{2y^2}{b^2}\right]^2 = 0 \\ \text{or, } & \left(\frac{x}{a} - 1\right) + \frac{2y^2}{b^2} = 0 \\ \text{or, } & \frac{2y^2}{b^2} = -\left(\frac{x}{a} - 1\right) \\ \text{or, } & y^2 = -\frac{b^2}{2a}(x - a) \end{aligned}$$

This represents the equation of a parabola, with vertex at $(a, 0)$.